

SAAKSHI - High-Level Design

Technical proposal · Gujarat Police Innovation Challenge 2026 · Model 1
(compulsory) + Hybrid

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Technical proposal for the Gujarat Police Innovation Challenge 2026. Submitted under **Model 1 (compulsory) + Hybrid**, as sanctioned by the problem statement's explicit permission for "a hybrid architecture combining features from two or more reference solution models".

Every number in this document is traceable. docs/claims-provenance.md lists each printed figure against the ticket that measured it and the command that reproduces it. The sizing and cost tables here are **computed at build time** by the same computeSizing() model that runs behind the console's /sizing screen, so this document and the product cannot disagree. Rebuild with `npm run docs:render`.

What this document is, and what it refuses to claim

A statewide CCTV analytics platform is easy to describe and hard to believe. In a field where every proposal promises 95% accuracy and seamless integration with systems the proposer has never been given access to, the only useful thing a design document can do is be **checkable**.

So this HLD states, up front, the five things SAAKSHI does **not** do:

- **No live VAHAN / SARTHI / eGujCop / AFIS / NAFIS connection.** The connectors are specified and a mock provider ships behind the same interface. We do not claim access we were never given.
- **No face recognition, and no biometrics of any kind are processed or stored.** Deliberate, and the reasoning is in §5.
- **No central video storage.** The architecture refuses it, and §3 is the arithmetic for why.
- **No VLM "suspicious activity detection."** Unfalsifiable, unauditable, and unaffordable at 80,000 cameras. §5 gives all three reasons.
- **No accuracy claim without a measurement.** Our measured ANPR read accuracy **misses the challenge's >90% detection/processing accuracy target, and is reported as missing**. That figure is printed in §5, on the deck, and in the government-feed output report, identically, because a system that hides the row it fails cannot be trusted on the rows it passes.

Everything that follows is designed to be re-run by a sceptical reader.

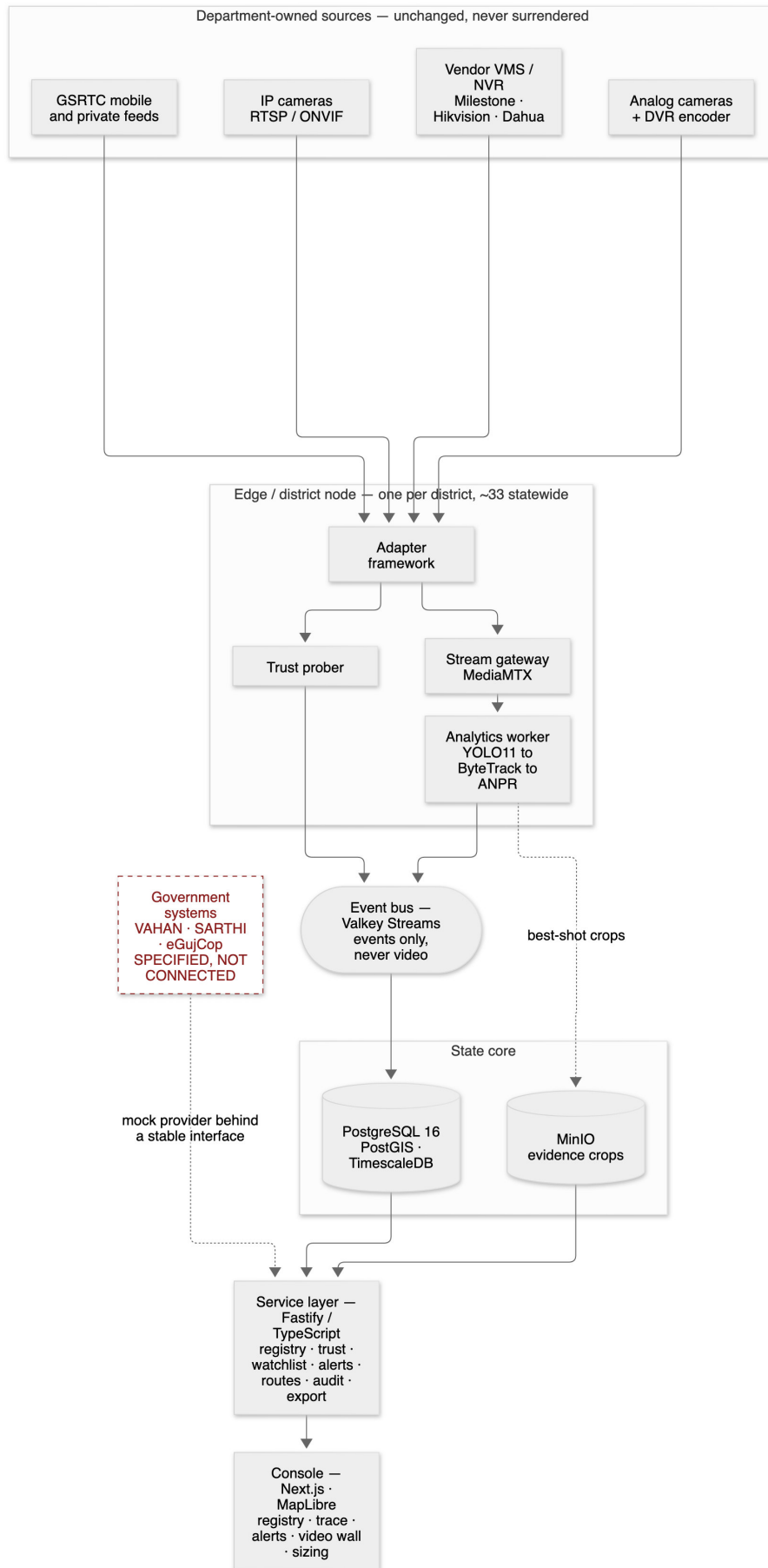
Requirement checklist

The problem statement lists what the High-Level Design must cover. Each row names the section that answers it.

#	Required content	Section
R1	Overall solution architecture with high-level diagrams and component interactions	§1
R2	Approach for integrating heterogeneous cameras, NVRs and VMS into a unified platform (IP, analog, multi-vendor, varied protocols)	§2
R3	Architecture for ingesting, processing and managing live streams from geographically dispersed locations	§3
R4	Approach for integrating live feeds with watchlist databases and continuously correlating analytics results to generate real-time alerts	§4
R5	AI-powered analytics approach: ANPR (mandatory), object detection, person/vehicle tracking	§5
R6	Alert generation and notification workflow: prioritisation, visualisation, user interaction	§6
R7	Scalability, interoperability, security and performance for statewide deployment	§7
R8	Technical prerequisites, assumptions, and information required from participating departments	§8
R9	Assumptions and constraints	§9

1. Overall solution architecture, with high-level diagrams and component interactions

SAAKSHI is a **metadata platform, not a video platform**. Cameras stay where they are, owned by the department that owns them; video stays inside the district it was captured in; and what crosses the state network is a stream of structured events a few hundred bytes wide. Every architectural decision below follows from that one choice, and §3 shows the arithmetic that forces it.



System context

Figure 1 - System context: sources, edge, core, console.

The four layers, and what each is responsible for

Layer	Responsibility	Technology	Scales by
Sources	Nothing. Unchanged, unmanaged by us, still owned by the department	Existing IP, analog+DVR, vendor VMS	-
Edge / district node	Decode, detect, track, read plates, probe health. Emit events, never video	Python 3.11, OpenCV, YOLO11, ByteTrack, ONNX OCR, MediaMTX	One node per district; streams per node
State core	Durable state, correlation, alerting, evidence, audit	PostgreSQL 16 + PostGIS + TimescaleDB, Valkey Streams, MinIO	Read replicas, consumer groups, erasure coding
Service + console	API surface and the operator's day	Fastify (TypeScript strict), Next.js 15, MapLibre	Stateless replicas behind a load balancer

Component interactions that matter

- **Adapter -> gateway -> analytics.** The adapter normalises a heterogeneous source into one internal stream contract (§2). Nothing downstream of the adapter knows what vendor it came from.
- **Analytics -> event bus.** The worker writes sightings, plate reads and health observations to Valkey Streams. It never writes to the database, so a slow database cannot stall a decoder.
- **Bus -> API.** Consumer groups give at-least-once delivery with replay; a district that loses backhaul buffers locally and drains on reconnect rather than losing the window.
- **API -> console.** REST plus server-sent events. The alert queue is pushed, not polled.
- **Evidence is never public.** A crop reaches a browser as a same-origin relative path, /evidence/crop?url=..., streamed by the API behind the session cookie and a role check. The object store has no public domain and hands out no shareable link.

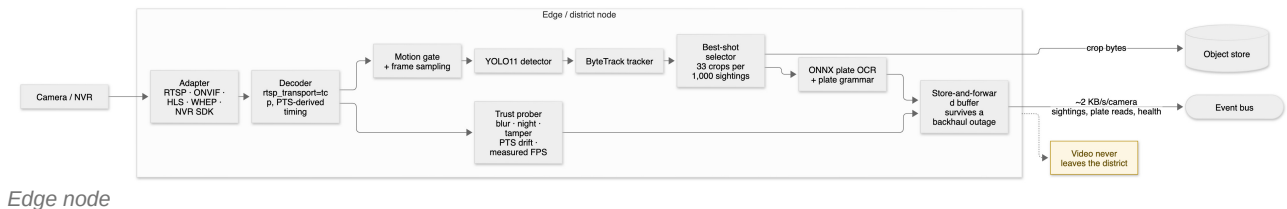
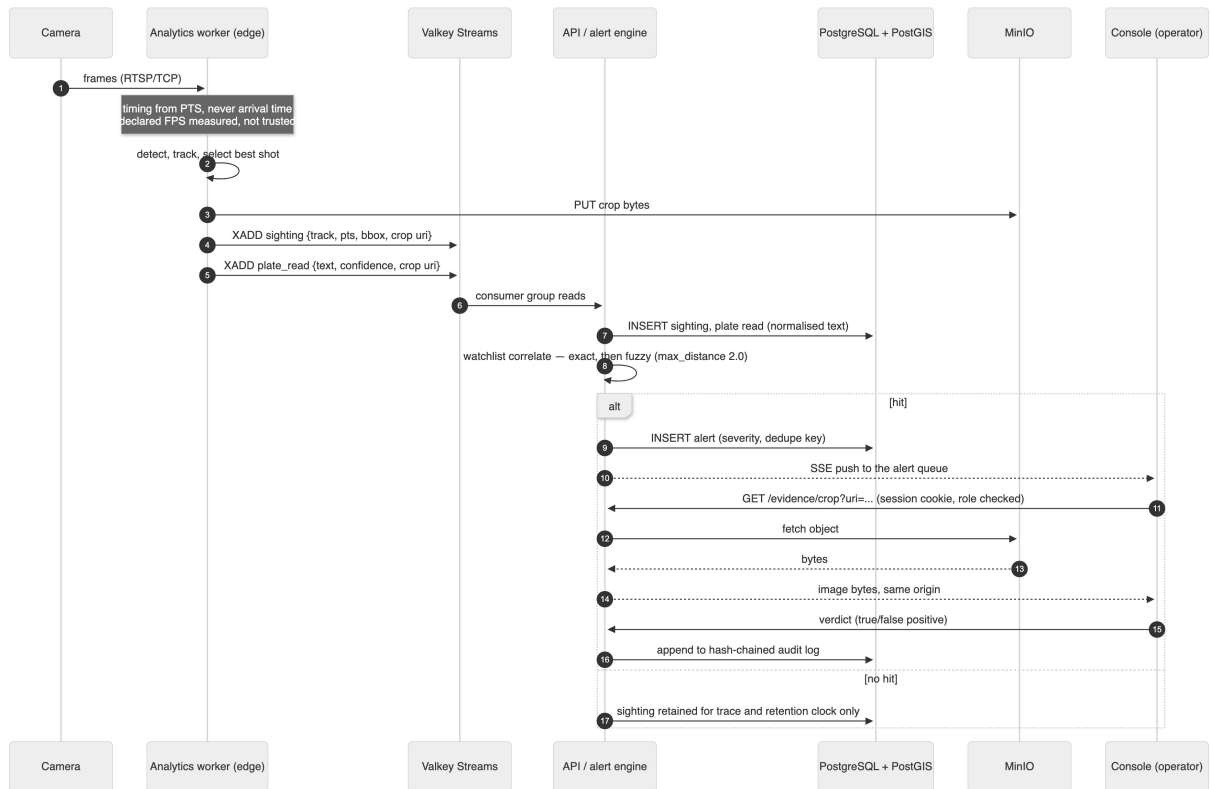


Figure 2 - Inside one edge / district node.



Data flow

Figure 3 - Data flow: frame to sighting to alert to evidence.

2. Approach for integrating heterogeneous cameras, NVRs and VMS into a unified platform

Gujarat's estate is not one system. It is ~80,000 cameras across **26 departments**, bought over a decade, under live AMCs, in a mix of IP and analog, behind at least four vendor VMS products. Any design that requires those departments to replace, re-cable or surrender their equipment is a design that will not be deployed, whatever its merits.

So SAAKSHI integrates at the protocol boundary and asks for nothing else. Full detail: `docs/adapter-framework.md`.

The adapter contract

Every source type is a driver behind one interface - `probe()`, `open()`, `frames()`, `close()` - returning the same normalised stream regardless of origin.

Source	How it is reached	Notes
IP camera	RTSP, forced over TCP	UDP loses packets on a congested WAN and produces corrupt frames that look like analytics failures
IP camera, discoverable	ONVIF Profile S	Device and media service; capabilities read, never assumed
Analog camera	Existing DVR/encoder's RTSP or HLS output	No re-cabling: the encoder the department already owns is the integration point
Vendor VMS / NVR	Vendor SDK or the VMS's own RTSP re-stream	Re-stream preferred - an SDK is a procurement dependency, a re-stream is not
Low-latency web	WHEP / WebRTC	Used by the console's video wall
Recorded / VOD	HLS	The challenge sandbox itself is VOD HLS

Three rules learned the hard way, and they are in the code

1. **Timing comes from PTS, never from frame arrival time.** A gateway that replays a buffered GOP on connect makes an arrival-time tracker compute impossible velocities after every reconnect - which then surface as false "impossible transition" alerts. Presentation timestamps are the only defensible clock.
2. **Never trust declared FPS.** `CAP_PROP_FPS` is what the device claims. We measure the real rate and store both, and **the delta is a product feature**: a camera declaring 25 fps and delivering 4 is a camera whose evidentiary value is a quarter of what the register says.
3. **Reconnect with backoff, 2 s -> 30 s, and treat join-time decoder warnings as normal.** A tight reconnect loop against a struggling NVR takes down the NVR, and then the department.

Interoperability, stated as obligations we accept rather than features we claim

- Adding a vendor is **one driver class**, not a schema change or a redeployment.
- The registry API is OpenAPI-documented (`docs/registry-api.md`) so a department can bulk-register its own estate without us.
- Everything in the stack is open source, which the challenge's About page states solutions *should* use. The single proprietary option - the natural-language query LLM - sits behind a `QueryCompiler` interface with four providers, one of which is local `ollama` and one of which is none. **Nothing proprietary is load-bearing.**

3. Architecture for ingesting, processing and managing live streams from geographically dispersed locations

The arithmetic that decides the architecture

This is the central engineering claim of the submission, and it is checkable in one line.

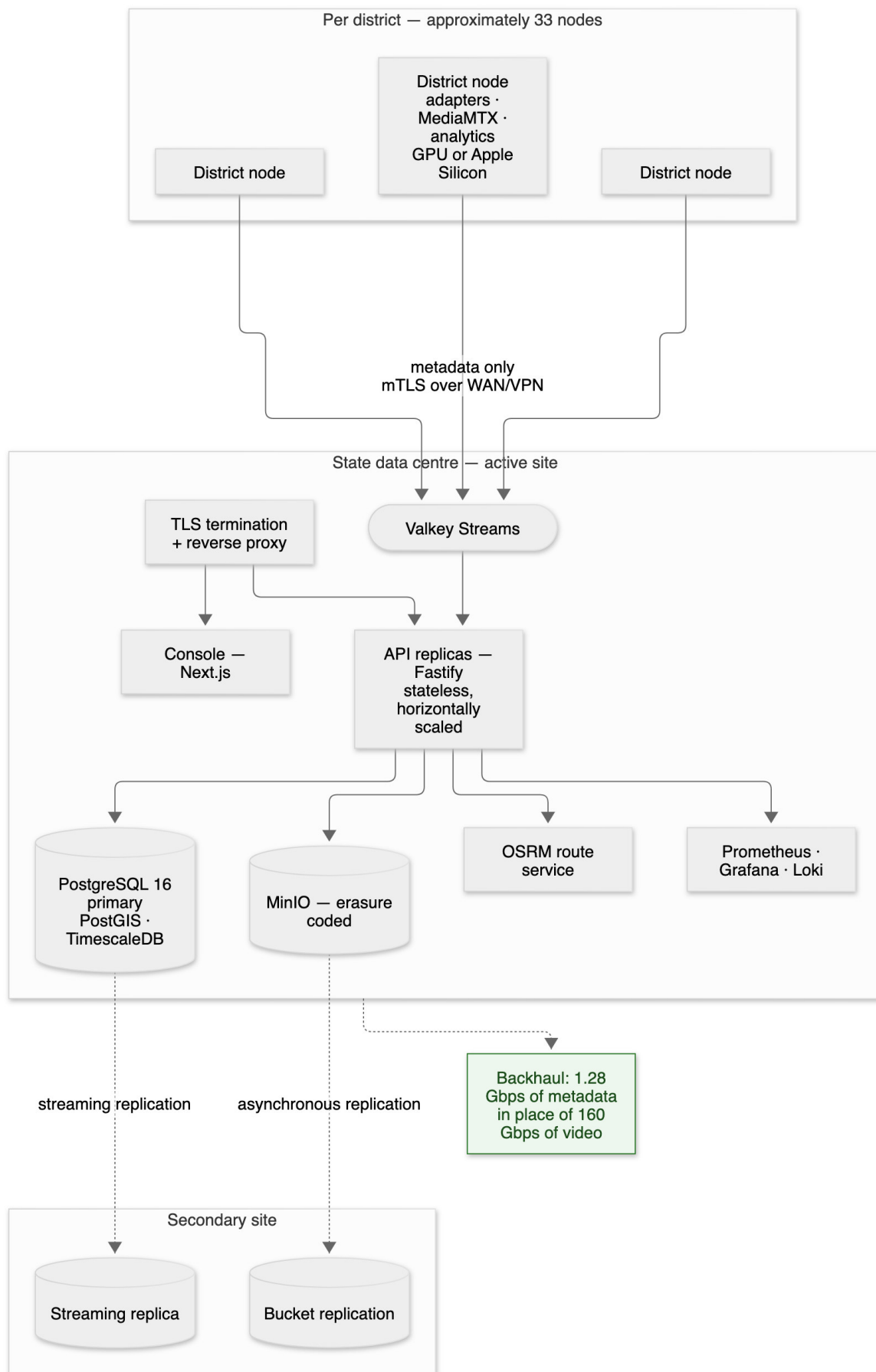
Every camera streamed to a central VMS - 80,000 x 2 Mbps	160.00 Gbps
Metadata only, from cameras analysed at the edge	1.28 Gbps
Total under this architecture	1.28 Gbps
Reduction	125x

We quote 125x, which is the smaller of two defensible numbers. It runs on the original assumed 2,000 B/s event rate and therefore reproduces PROJECT .md §9 exactly, so a reader can check it against a published section. Our *measured* per-track event rate takes the same estate to 2848x. The measured rate makes the architecture look better, not worse, and we lead with the figure that can be verified rather than the figure that flatters.

160 Gbps of sustained ingest is not a budget line; it is a reason a project fails. That is why a fully centralised VMS - Model 4 as /problems describes it - is rejected in this document with its arithmetic attached.

Managing streams across geographically dispersed locations

Concern	Design response
Unreliable district links	The node buffers locally and drains on reconnect. A backhaul outage delays events; it does not lose them
Bandwidth asymmetry	Only events cross the WAN. A crop is a few kilobytes and only for a best shot
Clock skew across districts	PTS-derived event time plus server receipt time, both stored. Correlation uses the former
Feeds that loop or cut	A hard scene cut is normal in this estate. Track IDs and galleries reset at a cut rather than bleeding across it - otherwise one vehicle inherits another's route
Node loss	District nodes are stateless between events. A replacement node re-registers and resumes; nothing is pinned to a machine
Central saturation	Consumer groups scale horizontally; the database takes writes in batches, and TimescaleDB partitions the sighting hypertable by time



Deployment topology

Figure 4 - Deployment topology for statewide operation.

Edge versus centralised - the decision, stated plainly

Analytics at the edge, state at the centre. Detection, tracking and plate reading run on the district node; correlation, watchlists, alerting, evidence and audit run centrally. Video never leaves the district. This is the only split that survives the 160.00 Gbps figure above, and it is also the split that keeps a department's footage under that department's control - which is what makes 26 departments willing to participate at all.

4. Approach for integrating live feeds with watchlist databases, and continuously correlating analytics to generate real-time alerts

What is built, and what is specified - the distinction is the point

There is no live VAHAN, SARTHI, eGujCop, AFIS or NAFIS connectivity in this submission, and this document never implies otherwise. We were not granted access to those systems, and a demonstration against a system we cannot reach would be a fabrication.

What exists instead is the part that is actually ours to build: a **watchlist service behind a provider interface**, with a mock provider that implements it, a documented connector specification for each government system, and the correlation engine - which is real, running, and indifferent to where the watchlist rows came from. Swapping the mock for a live VAHAN client is one class implementing one interface. Full detail: docs/watchlist-integration.md.

Watchlist category named in the problem statement	Supported	Source when live
Stolen vehicles	Yes, by registration number	VAHAN / state stolen-vehicle register
Blacklisted vehicles	Yes	Department-maintained list, bulk-loaded
Suspect watchlists (vehicle-linked)	Yes	eGujCop case linkage
Wanted persons	Vehicle-linked only	Correlated through a registered vehicle, never by face
Missing persons	Vehicle-linked only	Same

Wanted and missing persons are matched through vehicles, not through faces. That is a real limitation and we state it rather than let a category tick imply a capability. See §5.

Continuous correlation

Every plate read is correlated as it lands, not on a batch schedule:

1. **Normalise.** The read is folded to a canonical form on the write path, so the string that alerts is the same string a trace later searches.
2. **Exact match** against the active watchlist.
3. **Confusion-aware fuzzy match** if exact fails. Indian plates confuse a specific, known set of character pairs under motion blur and at night, and the matcher is weighted for those pairs rather than using a generic edit distance. The operating point is max_distance **2.0** - measured at **99.9% recall and 100% precision** at $d \leq 2$ (docs/fuzzy-matching.md §6).
4. **Emit an alert** with a severity and a dedupe key, or retain the sighting for trace and the retention clock only.

Fuzzy matching is the single highest-leverage feature in the system on this estate, because on this estate most plates are not read cleanly. A matcher that only accepts exact strings converts a marginal read into a miss.

5. AI-powered analytics approach: ANPR, object detection, and vehicle tracking

ANPR - the only mandatory analytic, and our measured numbers

Pipeline: motion gate -> YOLO11 detection -> ByteTrack tracking -> best-shot selection -> ONNX plate OCR -> plate-grammar normalisation -> watchlist correlation. Best-shot selection is what makes it affordable: OCR runs on the single best frame of a track, not on every frame of it, and the measured keep rate is **33 crops per 1,000 sightings**.

The measurement, on this estate, day and night sampled separately - 120 vehicle instances hand-labelled, method in docs/anpr-accuracy.md §2-3:

Metric	Measured
Exact read recall (reads equal to the human label, over legible plates)	0 of 3 - 0%
Precision (correct reads over all reads emitted)	0%
Plate-detection recall (plate boxes over human-legible plates)	100% on n=3
Character accuracy (1 ? editDistance/len)	51.8%
Human-legible plates in the hand-labelled sample	3 of 120
Against the challenge's >90% target	MISSES the challenge's >90% detection/processing accuracy target, and is reported as missing

Why, and why it is reported rather than buried. Of 120 hand-labelled vehicle instances on the sandbox estate, **three carry a plate a human can read at all**. The cameras are mounted for scene overview, not for plate capture: wrong height, wrong angle, wrong focal length, and at night a measured luma floor that leaves no plate signal to recover. The highest-confidence read of the government-feed run - **0.888** - is a roadside hoarding's phone number, not a plate.

This is the most important number in the submission, and it is a failure. It is also a **camera placement finding, not a model finding**, which is precisely the finding a statewide gap analysis exists to produce: the system correctly reports that this estate cannot do ANPR, and identifies the **24,000 cameras** in the statewide model that would need to be ANPR-viable for it to. A vendor claiming 95% on these feeds is claiming something the feeds cannot physically support.

The other analytics

Analytic	Status	Honest position
Object detection (vehicle, person, two-wheeler)	Running	YOLO11; the detector is the reliable half of the pipeline
Multi-object tracking	Running	ByteTrack. Measured caveat: 16 of 75 tracked passes (21%) do not hold a single vehicle - 9 identity switches, 7 unadjudicable. Anything treating one track as one vehicle inherits that rate
Camera trust scoring	Running	Classical CV, deterministic and explainable. §7 and Figure 5
Route reconstruction	Running	OSRM over a 540,711-way Gujarat road graph; 20-sighting trace builds in 125 ms p95 against a 3,000 ms budget
Impossible-transitions / plate-cloning detection	Running	A worked example: two reads 9.24 km and 30 s apart imply 1,109 km/h
Vehicle re-identification	Built, measured at 0.761 precision, ships DISABLED	Below the 0.900 bar. Roughly one appearance link in four would be wrong, and a wrong link attaches another vehicle's movements to this vehicle's evidentiary route. Off in three independent places (docs/limitations.md)
Face recognition	Deliberately out of scope	See below

Face recognition is deliberately out of scope

SAAKSHI processes no biometrics and stores no biometric template. Three reasons, in order of weight:

1. **It is not mandated.** ANPR is the only analytic the challenge makes compulsory. FRS is not asked for, so shipping it would be scope we awarded ourselves.
2. **It requires separate legal authorisation.** Face recognition against a public estate is a different legal instrument from vehicle analytics, with a different authority, a different retention regime and a different consent position. A hackathon submission is not the venue to assume that authorisation exists.
3. **It would poison the evidentiary argument.** This system's value is that every output is auditable: a plate read carries a timestamped crop, a confidence, and a hash-chained record of who looked at it and why. A face match carries the same weight in a courtroom as *"the model said so"*, and mixing the two lowers the standard of the whole platform.

The same reasoning rejects **VLM-based "suspicious activity detection"**: *suspicious* is not a definable class, so accuracy cannot be measured and any claim about it is unfalsifiable; it is unauditable as evidence; and the inference cost and latency at 80,000 cameras do not survive a procurement review.

6. Alert generation and notification workflow: prioritisation, visualisation, user interaction

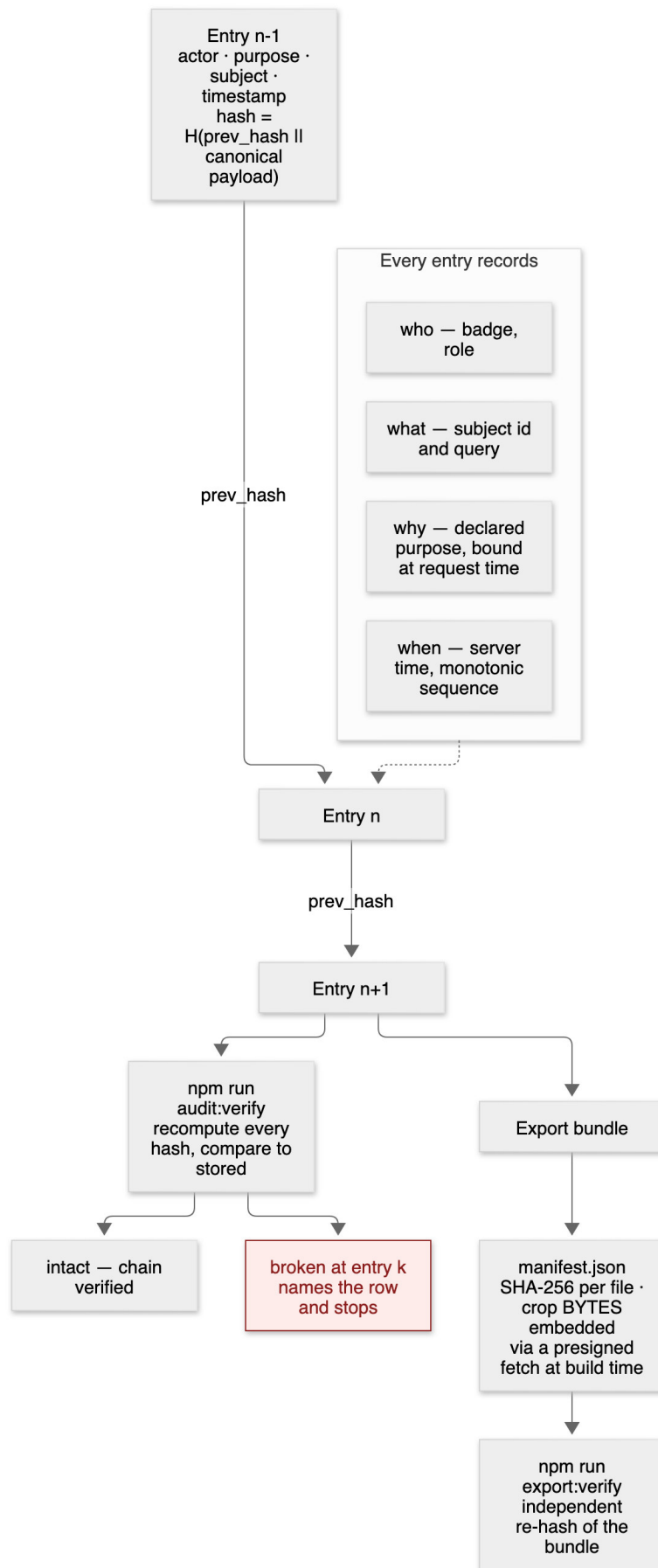
A control room does not fail because it has too few alerts. It fails because it has too many, and the operator stops reading them. The alert design is built around that failure mode. Full detail: [docs/alerting.md](#).

Prioritisation

Mechanism	What it does
Severity	Derived from the watchlist category and match confidence, not from the analytic that fired
Deduplication key	One vehicle passing one camera repeatedly is one alert with a count, not forty
Match-type banding	An exact match and a fuzzy match at d=2 are visibly different things in the queue; the operator is never asked to guess which they are looking at
Camera trust weighting	An alert from a camera the trust probe scores poorly is presented as weaker evidence, because it is weaker evidence

Visualisation and interaction

- **The alert queue is pushed over SSE**, not polled; a verdict round trip measured at **132 ms**.
- Every alert opens with **the crop that caused it**, the camera, the PTS-derived time, the confidence, and the matched watchlist entry. An alert a human cannot adjudicate in five seconds is an alert that will not be adjudicated.
- **The operator's verdict is the product**. True positive / false positive is one click, it is recorded against the alert, and it appends to the audit chain. That verdict stream is what makes measured precision possible at all - without it, nobody knows what the system is doing.
- **Escalation to a trace**. From an alert, one click reconstructs the vehicle's route across the estate, with each sighting's evidence and its confidence shown rather than smoothed away.
- **Export**. An evidence bundle embeds the crop **bytes**, a `manifest.json` with a SHA-256 per file, and is independently re-verifiable with `npm run export:verify`.



Audit chain

Figure 6 - The tamper-evident audit chain and export bundle.

7. Scalability, interoperability, security and performance for statewide deployment

Sizing - computed, not asserted

The two figures the challenge publishes disagree: /problems says **~80,000 cameras**, and /evaluation-criteria says **1,00,000+ records/endpoints**. We design to the higher figure and state both sources. The benchmark column below is the 1,00,000-camera case.

Every cell is produced at build time by computeSizing() - the model behind the console's /sizing screen - from constants each tagged *measured*, *vendor-listed* or *assumed* in docs/sizing-model.md §6.

	Pilot (500)	Statewide (80,000)	Benchmark (1,00,000)
ANPR cameras	500	24,000	30,000
Accelerators required	63	240	300
District nodes	33	33	33
Events per day	1,05,84,000	1,69,34,40,000	2,11,68,00,000
Retained storage	2-7 TB	841-3493 TB	1052-4367 TB
Total backhaul	0.00 Gbps	0.06 Gbps	0.07 Gbps
Capex	INR 3.07 crore - INR 5.12 crore	INR 11.70 crore - INR 19.50 crore	INR 14.63 crore - INR 24.38 crore
Annual opex	INR 49.52 lakh - INR 83.46 lakh	INR 2.23 crore - INR 7.05 crore	INR 2.85 crore - INR 8.93 crore
Annual cost per camera	INR 22,189 - INR 37,168	INR 571 - INR 1,369	INR 578 - INR 1,380

Costs are **bands, not points**, everywhere. A crop size measured on small replay frames is honestly 3-15 KB per crop until one live-feed measurement narrows it, and a model that hid that uncertainty behind a single number would be lying with more decimal places.

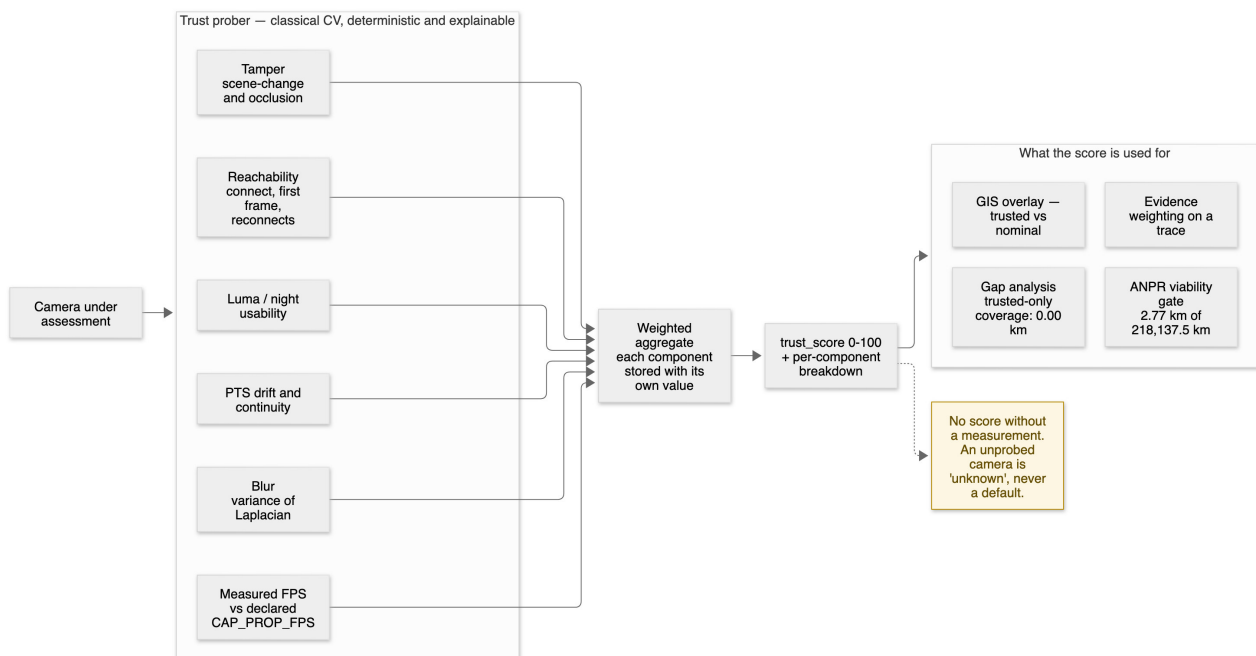
Performance - the six stated benchmarks, measured

Two of the six are misses and one is not meaningfully measured. All three are printed here.

Target	Measured	Verdict
1,00,000+ camera records	1,00,000 cameras benchmarked	Meets
API response < 200 ms	p95 110 ms at 200 concurrent; 252 ms at 500	Meets to 200 concurrent; over target at 500
Dashboard load < 3 s	readable 132 ms ; map fully drawn 1.65 s	Meets
Detection accuracy > 90%	exact read recall 0%	MISSES
Uptime > 99%	100.000% over a stable 30 min; 73.5% over a build hour	Not measured over a meaningful period
500+ concurrent users	500 concurrent, zero failed responses	Meets on throughput; p95 over target

Security and access control

Control	Implementation
Authentication	Badge + password to the web BFF, which holds an httpOnly session cookie. The browser never holds a bearer token
Authorisation	Role matrix - administrator, supervisor, operator, auditor - enforced at the route, not in the UI (docs/rbac.md)
Purpose binding	A sensitive query declares <i>why</i> , and the declared purpose is recorded with the query. A search with no stated purpose is refused
Evidence access	Crops stream through the API behind session and role. The object store has no public domain, and a crop URL is not shareable - it requires a live session, which is the point
Tamper-evidence audit	Hash-chained log, each entry over the previous hash. <code>npm run audit:verify</code> recomputes the whole chain and names the row it breaks at
Export integrity	Bundle manifest with SHA-256 per file, re-verifiable independently
Retention	An evidence clock per record; the retention window on this estate is 7-15 days and expiry is enforced, not advisory
Transport	TLS at the edge of the state network; district -> core over mTLS



Trust-score pipeline

Figure 5 - The trust-score pipeline.

Registry completeness - the differentiator, and a number nobody enjoys printing

A registry that lists cameras is a spreadsheet. A registry that states which cameras can be **relied on** is infrastructure. The trust probe measures reachability, real versus declared FPS, blur, night usability, tamper and PTS continuity, and the gap analysis then asks the only question that matters operationally: *how much of Gujarat's road network is actually covered by a camera you would take to court?*

Road network analysed	540,711 ways · 218,137.5 km (Geofabrik western zone clipped to Gujarat, OSM data 2026-09-05)
Covered by any camera	21.47 km - 0.0098%
Covered by an ANPR-viable camera	2.77 km - 0.0013%
Covered by a trusted camera	0.00 km
Junctions with zero trusted coverage	6,750 of 6,750

The trusted-only figure is 0.00 km, a 100% delta against apparent coverage. On the sandbox estate, no camera clears the trust bar. That is the finding. A procurement officer holding this report knows exactly which cameras to fix first and why - which is worth considerably more than a dashboard that reports 30 cameras online and stops there.

8. Technical prerequisites, and the information required from participating departments

Deployment is blocked by information, not by engineering. This section is the ask, so a participating department can answer it before a single node is racked.

Technical prerequisites

Prerequisite	Why it is needed	If it is absent
One node per district: GPU or equivalent accelerator, per the §7 sizing	Analytics run at the edge	Fewer concurrent ANPR streams per district; the model computes the trade directly
Network reachability from that node to each camera or NVR, RTSP/TCP	Ingest	Per-camera exclusion, recorded in the registry rather than silently skipped
WAN link, district -> state core, sized from §7's metadata figure	Event backhaul	Local buffering absorbs outages; sustained under-provision delays alerts
Outbound TLS/mTLS from district to core	Transport security	Deployment blocked
A camera location for every camera - latitude and longitude	GIS mapping and gap analysis are compulsory Model 1 deliverables	A camera with no coordinates is unassessable and is reported as such, never quietly dropped
NTP on every node	Cross-camera correlation	Skew produces false impossible-transition alerts

Information required from participating departments

#	What we need	Why
1	Camera inventory: id, site name, lat/long , mounting height, view direction	The registry and every spatial output depend on it
2	Stream endpoints and protocol per camera (RTSP/ONVIF/HLS), and credentials	Ingest
3	VMS/NVR make, model and version where a VMS is in front of the cameras	Selects the adapter driver
4	Which cameras are intended for plate capture	Sets the ANPR-viability baseline against what we measure
5	Retention policy and legal retention window per department	Drives the evidence clock; 7-15 days observed on this estate
6	Named authority for watchlist provisioning, and its update cadence	Correlation is only as current as its watchlist
7	Operator roster with roles, for the RBAC matrix	Access control
8	Declared lawful purposes for sensitive queries	Purpose binding is enforced, so the list must exist
9	AMC status and contact per camera estate	A camera nobody can repair is a permanent gap, and the gap analysis should say so
10	Escalation path for an alert a department owns	An alert with no owner is an alert nobody actions

D4-06 expands this into a standalone department onboarding questionnaire. This section is self-contained and complete as it stands; the questionnaire is the same content as a form a department fills in.

9. Assumptions and constraints

These are accepted limitations carried from the project's running defect and finding log (BL - 01), triaged before submission. They are stated here because a design document that lists only its strengths tells a reader nothing they can use.

Assumptions

#	Assumption	If it is wrong
A1	Departments will expose a stream endpoint but will not surrender their video	The architecture is unchanged - this is the assumption it is built on
A2	One accelerator-equipped node per district is procurable	The sizing model recomputes for any node count; coverage per district falls
A3	Camera coordinates can be supplied or surveyed	Spatial outputs degrade to a table view; cameras are marked unassessable, not guessed
A4	2 Mbps per camera is representative for the centralised comparison	Vendor-listed, tagged as such in the sizing model; the ratio moves, the conclusion does not
A5	Crop size on live feeds lands in the 3-15 KB band	Measured only on small replay frames. Every storage figure is a band because of it
A6	The measured event rate from the sandbox generalises	Tagged <i>measured</i> with its ticket. One quiet-machine re-measurement is the stated caveat

Constraints and accepted limitations

#	Constraint	Status
C1	ANPR read accuracy on this estate is 0%	Accepted and published. A camera-placement finding, not a model finding (\$5)
C2	No live government-database connectivity	By design. Connectors specified, mock provider ships (\$4)
C3	No face recognition; no biometrics processed	Deliberate (\$5)
C4	Vehicle re-ID ships disabled at 0.761 precision	Below the 0.900 bar; three independent off-switches (docs/limitations.md)
C5	Cross-camera re-ID is unmeasured, and the diagnostic points the wrong way	No plate anchor exists on this estate to label a cross-camera pair. Stated, not estimated
C6	21% of tracked passes do not hold a single vehicle	Measured. Anything treating one track as one vehicle inherits it (\$5)
C7	Uptime is not measured over a meaningful period	100.000% over 30 stable minutes is not a production availability claim, and is not presented as one
C8	API p95 exceeds 200 ms at 500 concurrent	110 ms at 200, 252 ms at 500. Reported as a partial meet
C9	Impossible-transition sweep flagged 2 of 7 assessable transitions	The tool itself flags that rate as implausibly high for genuine cloning - a sparse-estate artefact, reported rather than presented as 28.6% cloning
C10	Trusted road coverage is 0.00 km	The estate's finding, and the reason the gap analysis exists (\$7)
C11	Analog cameras are reached only through an existing encoder	We add no hardware. A department with no encoder needs one before integration
C12	The NL-query LLM is the one optional proprietary dependency	Behind a four-provider interface including local ollama and none. Nothing proprietary is load-bearing

docs/limitations.md carries the measured detail behind C4-C6, and D4-08 triages the full BL - 01 log into

limitations and a roadmap.

How to check this document

Nothing here asks to be taken on trust.

```
npm run docs:render           # regenerates this document and its PDF
npm run docs:linkcheck        # every internal link resolves
npm run deck:build            # the deck, from the same sizing model
npm run report:gap-analysis    # the coverage figures in section 7
npm run audit:verify          # recompute the whole audit chain
npm run export:verify         # re-hash an evidence bundle independently
npm run test -w @saakshi/api -- hld    # asserts this document against the deck
```

Document	What it holds
docs/claims-provenance.md	Every printed figure, its ticket, and the command that reproduces it
docs/sizing-model.md	The sizing model and every constant's provenance tag
docs/adapted-framework.md	§2 in full
docs/watchlist-integration.md	§4 in full
docs/alerting.md	§6 in full
docs/anpr-accuracy.md	The §5 measurement's sampling method
docs/trust-score.md	The §7 trust pipeline
docs/chain-of-custody.md	The audit chain and export manifests
docs/limitations.md	Measured limitations
docs/deployment.md	How it is deployed and run

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